

Question Number	Answer	Additional Guidance	Mark
1(a)	A description that contains five of the following: • clear statement of the dependent variable (1) • use of five different concentrations (1) • plants set up with suitable apparatus (to enable counting leaves/collecting plants) (1) • For a (stated/same) time • identification and control of one variable (1) • repeats and calculate means (to compare) (1) • description of calculation of rate (1)	Eg (counting) number/mass of plants/leaves Ignore rate. Units not needed Eg petri dish/beaker (so plants can be counted/find mass) not test tube minimum 1 day Accept: Temperature – thermostatic water bath /AC/incubator pH – buffer light intensity – bulb at set distance eg to find variability/SD not mean rate eg number/mass of plants/leaves/leaflets divide by time	(6)

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1(b)	An explanation that includes three of the following: • nitrogen used to form chlorophyll (1) • to form {amino acids/ proteins} (1) • (for) enzymes (1) • to form {nucleotides / RNA / DNA/ribosomes} (for the synthesis of chromosomes) (1) • to produce a named {protein/ organelle/membrane} involved in cell division (1)	Accept relevant named enzyme accept ATP/other organic compounds that can be formed e.g. tubulin/spindle fibres	(3)

(Total for Question 1 = 9 marks)

